An Evolution-Driven Computational Screen for RNA Motifs

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1 Introduction

1.1 Ribonucleic Acids

Central dogma of molecular biology: DNA $\rightarrow$ RNA $\rightarrow$ Protein.

In the hypothetical RNA world, RNA molecules are thought to have played a diverse set of roles in catalysis, regulation, and information storage. Although these tasks are primarily preformed by proteins and DNA in modern organisms, the RNA functions span beyond their roles as messengers.

RNAs form a variety of secondary and teriary structures, (like hairpins, pseudoknots, and G-quadruplexes, which can be involved in their functional roles.)

[maybe include figure of rna structures]

STRUCTURE LEADS TO FUNCTION

1.2 Functional Ribonucleic Acids

[Content here] Include background of main discoveries i.e. families of ribozymes, riboswitches, etc.

The diverse functional roles of RNA are driven by their structural and reactive diversity. Naturally occurring functional RNAs include riboswitches, and ribozymes, *in vitro* selected RNAs include flourogens and aptazymes (ligand triggered) (Micura & Höbartner, 2020).

RNA is involved in some of the most important recent biotechnological advancements. CRISR-Cas9, lncRNAs, mRNA vaccines, RNAi, etc.

RNAs involved in light, both as fluorescent aptamers and as photoreceptors (Micura & Höbartner, 2020). [Cite Andrej Luptak, e.g. photoriboswitch Werewolf-1 developed by Rotstan et al. (2020) using *in vitro* selection, and the RNA aptamer Spinach, which binds to a flourogen and becomes fluorescent (Paige et al., 2011)] These are important for... Medical research, biotech, agriculture, etc.

Many of these found in bacteria, but also in eukaryotes, and archaea. This project will focus on screening bacteria, but the methods could be applied to other domains of life.

WHERE ARE FUNCTIONAL RNAs FOUND - in 5' UTRs, introns, intergenic regions, etc. Riboswithces are often found in 5' UTRs, and regulate gene expression by changing their structure in response to ligand binding (Kavita & Breaker, 2023).

1.3 Current Methods for Functional RNA Discovery

Experimental approaches... e.g.

Bioinformatic approaches have had some success in identifying novel naturally occurring functional RNAs. These approaches typically rely on sequence conservation, structural conservation, or a combination of both. e.g. CMFinder

[Still very few found, so either they are very rare, or we are not looking in the right way.]

In vitro selection approaches have been used successfully to generate novel functional RNAs. These approaches typically involve the generation of a large library of random RNA sequences, which are then subjected to a selection process that enriches for sequences with a desired function (cite someone)

So why look for them naturally? - we would never have come up with the idea of CRISPR-Cas9, or mRNA vaccines, if we had only looked at *in vitro* selected RNAs. Its worth looking for novel families, not just more of the same, as valuble as these sorts of screens are - yeilded things like flouride riboswitches

1.4 Evolution based Bioinformatic Screens Yield Novel Functional RNAs

[Content here] Zasha Wienburg's lab has had successes with their tool CMFinder, Yao et al. (2006), developed in e.g. Sack et al. (2025) recently identified 16 new structured RNA classes in plants using an evolution based bioinformatic screen. This study focused on structured RNAs involved in alternative splicing.

Self cleaving riboswitches

1.5 Aims

The main aims of this project are:

1. Compile a list of candidate RNA motifs with potential regulatory or functional roles.
    - Multiple sequence alignments of 5' UTRs of mRNAs of specific gene families across multiple bacterial species.
    - Prediction of conserved RNA secondary structures in these alignments.
2. Create a reusable computational pipeline to systematically screen for these motifs.

1.6 Hypothesis

Evolutionarily conserved motifs in 5' UTRs of mRNAs of specific gene families will yield structural regulatory motifs.

1.7 Significance

References

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